HOW CAN IT BE THAT MATHEMATICS, BEING AFTER ALL A PRODUCT OF HUMAN THOUGHT WHICH IS INDEPENDENT OF EXPERIENCE, IS SO ADMIRABLY APPROPRIATE TO THE OBJECTS OF REALITY?

ALBERT EINSTEIN

"AND WHAT IS THE USE OF A BOOK," THOUGHT ALICE, "WITHOUT PICTURES OR CONVERSATIONS?"

LEWIS CARROLL (*ALICE IN WONDERLAND*)

. . . THE DESIGNER OF A NEW SYSTEM MUST NOT ONLY BE THE IMPLEMENTOR AND THE FIRST LARGE-SCALE USER; THE DESIGNER SHOULD ALSO WRITE THE FIRST USER MANUAL. . . IF I HAD NOT PARTICIPATED FULLY IN ALL THESE ACTIVITIES, LITERALLY HUNDREDS OF IMPROVEMENTS WOULD NEVER HAVE BEEN MADE, BECAUSE I WOULD NEVER HAVE THOUGHT OF THEM OR PERCEIVED WHY THEY WERE IMPORTANT.

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The overall goal of this chapter is to build our way up to vector spaces. Vector spaces are the mathematical structure that underpins quantum mechanics and, by extension, quantum communication. After the development of vector spaces, we briefly discuss some additional mathematical topics that are useful in quantum communication.

Group Theory

To begin our adventure towards vector spaces, we start with, what I consider, the most fundamental mathematical structure: groups. Groups are *sets*[1] equipped with an *operation* that satisfies four axioms. Groups are ubiquitous in mathematics and physics, and they have applications in many fields, including cryptography, coding theory, and particle physics.

[1] Briefly, a **set** is a collection of well-defined, distinct objects.

Operations

On any nonempty set A, an operation is a mapping from two elements in A onto some other, not necessarily unique, element in A. More formally,

Definition 1 (Operation)**.** A **binary operation** on a nonempty set A (simply, an **operation** on a set A), is a function $f : A \times A \to A$.

By this definition, A is closed under the operation defined by f.

We can familiarize ourselves with Defn. 1 by considering standard multiplication, denoted here by $(\cdot)$. Limiting ourself to the integers, $\mathbb{Z}$, we define $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ as $f(a, b) = a \cdot b$. One may repeat this process with integer addition $(+)$, defining $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ as $f(a, b) = a + b$. As a last example, suppose $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are vectors in $\mathbb{R}^3$. One way to define multiplication is the *vector product*, denoted $(\times)$, the mapping from two vectors to another vector, *i.e.*, $f : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ defined by $f\left(\vec{\mathbf{a}}, \vec{\mathbf{b}}\right) = \vec{\mathbf{a}} \times \vec{\mathbf{b}}$.

Later in the text we will drop the mapping label of f and simply use, *e.g.*, $(a, b) = \square$, where $\square$ is to be filled.

Question 2. *Given Defn. 1, why would we not consider the scalar product of vectors to be a binary operation?*

Groups

Consider the set of integers equipped with addition, often denoted $(\mathbb{Z}, +)$. From our past experience, we can list a couple of facts about this set:

1. the sum of two integers is still an integer, and
2. the order in which we add integers does not affect the outcome.

In symbols, the first fact is written as $\forall\ a, b \in \mathbb{Z}, a + b \in \mathbb{Z}$. This is known as as the *closure* property. Similarly, the second fact is written as $\forall\ a, b, c \in \mathbb{Z}$, $a + (b + c) = (a + b) + c$. This is known as *associativity*.

These two facts are things we have taken for granted since elementary school. If we now consider the integer 0, we can add two more facts to our list:

3. the sum of any integer and 0 is simply that integer, *i.e.*, $\forall\ a \in \mathbb{Z}$, $a + 0 = a = 0 + a$, and

4. there exists a *negative* for each integer such that the sum of an integer and its negative is 0, *i.e.*, $\forall\, a \in \mathbb{Z}$, $a + (-a) = 0 = (-a) + a$.

Mathematicians refer to any element for which property 3 holds as the *additive identity* (or the *identity element* in general) and $-a$ is known as the *additive inverse* of a.

Stripping the addition operator from the set and replacing it with the multiplication operator, we now have $(\mathbb{Z}, \cdot)$. We have enough experience to know that the first two facts still hold, *i.e.*, the product of two integers is still an integer and multiplication of integers is associative. However, the identity element in this case would be the integer 1 since, $\forall\, a \in \mathbb{Z}$, $a \cdot 1 = a = 1 \cdot a$. We encounter an issue when trying to satisfy criterion 4 from above. $\forall a \in \mathbb{Z}$, we seek an integer a^{-1} s.t. $a \cdot a^{-1} = 1 = a^{-1}a$. In the special case of $a = 1$, we simply choose $a^{-1} = 1$ and we satisfy the criterion. However, for $a = 2$, we would need to choose $a^{-1} = \frac{1}{2}$, not an integer! Thus, we cannot satisfy criterion 4 for all integers in $\mathbb{Z}$.

The fact that the *set* $(\mathbb{Z}, +)$ carries this additional property makes it *special*. When abstracting the four properties discussed above, we arrive at the following definition for the first mathematical structure we will study:

Definition 3 (Group)**.** A **group** is a nonempty set G equipped with a binary operation $(*)$ that satisfies the following axioms: $\forall\, a, b, c \in G$,

1. $a * b \in G$ (Closure)
2. $a * (\, b * c\,) = (a * b) * c$, (Associativity)
3. $\exists\, e \in G$ s.t. $a * e = a = e * a$, (Identity)
4. $\forall\, a \in G$, $\exists\, a^{-1} \in G$ s.t. $a * a^{-1} = e = a^{-1} * a$. (Inverse)

By this definition, the integers equipped with addition form a group. Note that in the previous definition we explicitly stated the closure property as an axiom even though this is already part of the definition of an operation.

Unlike the integers, the set of rational numbers excluding zero, denoted $\mathbb{Q} \setminus \{0\}$, do form a group when equipped with multiplication. To see this, consider generic $a, b, c, d \in \mathbb{Z}$ with $b \neq 0$ and $d \neq 0$. Then the values $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ (we assume the fractions here are in simplest form but this need not be true for what follows). Clearly, the product of these two values is still rational: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \in \mathbb{Q}$.[2] Associative multiplication in $\mathbb{Q}$ follows from the fact that multiplication of integers is associative. The identity element is the rational number 1 since

[2] The set of rational numbers requires that the denominator be nonzero. From previous experience, we know that the product of two nonzero integers is also nonzero. This property is consequential of the set of integers, $\mathbb{Z}$, being an integral domain. This is a topic we cover in the following section.

$\frac{a}{b} \cdot 1 = \frac{a}{b} = 1 \cdot \frac{a}{b}$. Finally, every rational number except zero has a multiplicative inverse, namely its *reciprocal*, *e.g.* $\frac{b}{a}$, s.t. $\frac{a}{b} \cdot \frac{b}{a} = 1 = \frac{b}{a} \cdot \frac{a}{b}$. Thus, by Defn. 3, $(\mathbb{Q} \setminus \{0\}, \cdot)$ is a group. It can also be shown that $\mathbb{R} \setminus \{0\}$, $\mathbb{Q}^+$, and $\mathbb{R}^+$ are all groups under multiplication.

We take a slight detour in our development of vector spaces to introduce an interesting application of groups in geometry. We start with the following definition:

Definition 4 (Symmetry group)**.** The **symmetry group** G of a mathematical object O is the set of all transformations that preserve the object's structure.

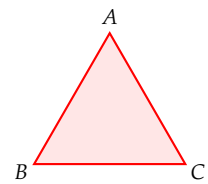


Figure 1: An equilateral triangle $\triangle ABC$. Note that the labels on the triangle's vertices are simply for us to visualize how the triangle is transformed. The labels do not imply that the triangle is oriented in any particular way.

For example, the symmetry group of an equilateral triangle $\triangle ABC$ (see Fig. 1) is the set of all transformations that preserve the triangle's shape, size, and *general* orientation. To build this group, we consider the following transformations:

1. Rotations: The triangle can be rotated (counterclockwise around its center) by $0°$, $120°$, or $240°$. Assigning names to each of these transformations, we define $\{e, r, r^2\}$, where e is the identity transformation (no rotation), r is the $120°$ rotation, and r^2 is the $240°$ rotation. We see each of these transformations in action below:

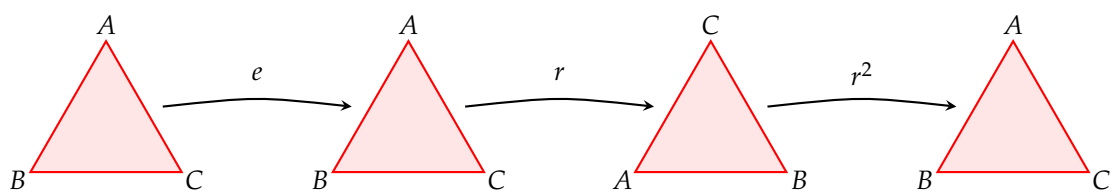


Figure 2: We see that the identity transformation e leaves the triangle unchanged, while the transformations r and r^2 rotate the triangle by $120°$ and $240°$ respectively. As one would expect, applying the r and r^2 rotations consecutively to the original triangle yields the original triangle again.

2. Reflections: Other than rotations around the centroid, we can also reflect the triangle about its axes of symmetry. The equilateral triangle has three axes of symmetry, each passing through one vertex and the midpoint of the opposite side. We can label these reflections as x, h, and v (see Fig. 4). We thus expand our set of transformations to $\{e, r, r^2, x, h, v\}$.

   Applying these new transformations to the original triangle in Fig. 1, we arrive at the results seen in Fig. 2.

As seen in Figs. 1–4, the transformations $\{e, r, r^2, x, h, v\}$ can be combined (*i.e.*, applied one after the other) to produce other, single transformations. If we think of a transformation as a function from

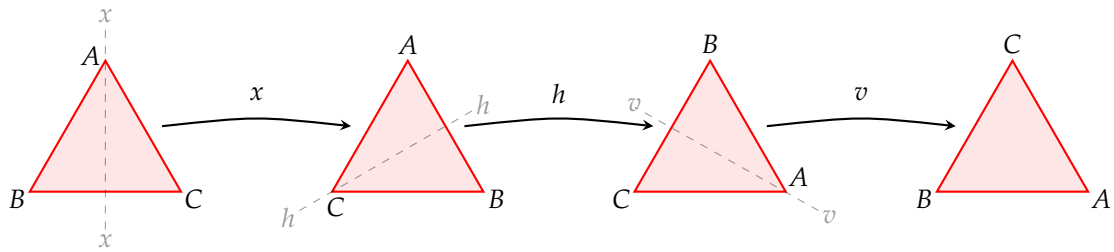


Figure 3: The axis of symmetry is shown (faintly) prior to reflection. Notice applying x, h, and v consecutively is equivalent to simply applying the h transformation to the original triangle.

the triangle onto itself, then the idea of combining transformations is equivalent to the composition of functions. In Fig. 2 (x followed by h and then followed by v), we can write $v \circ h \circ x$ (*i.e.*, first apply the x transformation, then the h, and finally the v) $= h$. In Fig. 1, we also saw that $r^2 \circ r = e$. Naming the set of transformations $D_3 \equiv \{e, r, r^2, x, h, v\}$[3] and equipping it with the composition operation, one can verify the Cayley table in Tab. 1.

[3] We reserve the symbol $\equiv$ to denote a definition. This is different from the symbol $\square \equiv \square \left(\text{mod}\, \square\right)$ used to denote congruence, which we will see in the next section.

Tab. 1 tells us that D_3 is closed under the composition operation, $(\circ)$, and composition of functions is known to be associative. Additionally, we've defined e as the identity element in this set and every element in D_3 has an inverse. For example, the inverse of r is r^2 since $r \circ r^2 = e = r^2 \circ r$. Similarly, the inverse of x is itself, *i.e.*, $x \circ x = e = x \circ x$. The same holds for h and v. Thus, we have shown that the set of transformations D_3 satisfies all four axioms in Defn. 3 and is therefore a group. In particular, it known as the **dihedral group** D_3, with the subscript of 3 indicating that our shape has 3 sides.

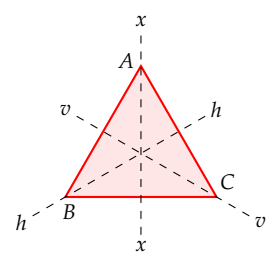


Figure 4: Showing all of the axes of symmetry on the equilateral triangle $\triangle ABC$.

In going through the above example, one may have noticed something rather peculiar about Tab. 1. Particularly, the fact the 3×3 block in the upper left corner is symmetric about the main diagonal, but no other 3×3 block in the table is symmetric. This symmetry tells us that, when applying rotations, the order in which we apply them does not matter. In other words, $r \circ r^2 = r^2 \circ r$, etc. This is not true for the reflections, however. For instance, $x \circ h \neq h \circ x$.

Groups that satisfy this additional property, the property of *commutativity*: $a * b = b * a$ for all $a, b \in G$, are called *abelian* groups. The name is derived from the mathematician Niels Henrik Abel, who made significant contributions to mathematics in the 19th century.

$\circ$	e	r	r^2	x	h	v
e	e	r	r^2	x	h	v
r	r	r^2	e	v	x	h
r^2	r^2	e	r	h	v	x
x	x	h	v	e	r	r^2
h	h	v	x	r^2	e	r
v	v	x	h	r	r^2	e

Table 1: The composition table for the symmetry group D_3 of the equilateral triangle. The way to read this table is as follows: the element in column i and row j is the result of applying the transformation i first and then the transformation j, *i.e.*, $j \circ i$. For example, the element in column r and row h is v, meaning that $h \circ r = v$.

Definition 5 (Abelian group)**.** A group is **abelian** if, $\forall\, a, b \in G$, it also satisfies the axiom

5. $a * b = b * a$. (Commutativity)

From what we mentioned in the previous paragraph, we see that the symmetry group D_3 is not abelian. However, the (proper) subgroup[4] of rotations, $\{e, r, r^2\}$, is abelian.

We can now return to the list at the start of this subsection and append this additional property:

5. the addition of integers is independent of the order of the integers (*i.e.*, it is *commutative*: $\forall\ a, b \in \mathbb{Z}$, $a + b = b + a$),

stating that $(\mathbb{Z}, +)$ is an abelian group. By extension of the properties of $\mathbb{Z}$, $(\mathbb{Q}, +)$ is also an abelian group. Thus, it then follows that $(\mathbb{Z}, +)$ is a proper subgroup of $(\mathbb{Q}, +)$.

[4] A nonempty subset H of a group G is a **subgroup** provided that it is closed and their exists an inverse for every element in the subgroup. This is true of the subset $\{e, r, r^2\}$ of D_3. We use the word 'proper' to say that the subgroup is not equal to the whole group D_3.

Question 6. *Produce the Cayley table of rotations, see Tab. 1, for the dihedral group D_4, the group of symmetries of the square.*

Further Resources

1. *Abstract Algebra: An Introduction* (3rd ed.) by Thomas W. Hungerford.

2. *Algebra: Abstract and Concrete* (ed. 2.6) by Frederick M. Goodman. https://homepage.divms.uiowa.edu/~goodman/algebrabook.dir/algebrabook.html

The material in this `\section{}` comes from Hungerford's text quite closely, though not in the same order. The structure is instead similar to Goodman's. Goodman does a thorough treatment of symmetries early in his text, which is why I have recommended it here. Goodman's text is also *free*[5] to download from his university webpage. Another (*free*) text that covers similar material is the following.

[5] See page ix of the text for its cost.

3. *A Crash Course on Group Theory* by Peter J. Cameron. https://cameroncounts.wordpress.com/wp-content/uploads/2016/11/groups.pdf

For a more traditional and advanced approach to group theory, consider the following text.

4. *Abstract Algebra* (3rd ed.) by I. N. Herstein.

Ring Theory

Rings are a natural extension of groups. Rings are sets equipped with two binary operations that generalize the arithmetic operations of addition and multiplication. Before we define rings formally, we first intrpoduce the concepts of *congruence* and *modular arithmetic*. These will not only be useful in our development of ring theory, but also in understanding the structure of rings themselves.

Congruence and Modular Arithmetic

Consider $a, b, c \in \mathbb{Z}$ with $c > 0$. We say **a is congruent to b modulo c** if c divides the difference of a and b. In symbols, $a \equiv b (\bmod n)$ if $c|(a-b)$.[6] For example, $15 \equiv 6 (\bmod 3)$ since 3 divides $15 - 6 = 9$. Additionally, $23 \equiv -1 (\bmod 6)$ since 6 divides $23 - (-1) = 24$.

Now, let $a, c \in \mathbb{Z}$ with $c > 0$. The **congruence class of a modulo c**, denoted $[a]$, is the set of all integers congruent to a modulo c, *i.e.*,

$$[a] = \{n : n \in Z \text{ and } n \equiv a (\bmod c)\}. \tag{1}$$

For example, in congruence modulo 7, $[5] = \{\ldots, -9, -2, 5, 12, 19, \ldots\}$.[7] Restructuring this, equivalently $[5] = \{5, 5 \pm 7, 5 \pm 14, \ldots\}$. Thus, Eqn. (1) can be expressed instead as follows

$$[a] = \{a + kc : k \in Z\}, \tag{2}$$

a more digestible version.

Compactly, the set of all congruence classes modulo n is denoted $\mathbb{Z}_n$, read "$\mathbb{Z}$ mod n."[8] One may anticipate an issue in terms of representatives of these classes. For example, since $4 \equiv 7 (\bmod 3)$, $4 \equiv 16 (\bmod 3)$, and $4 \equiv -2 (\bmod 3)$, it can be argued that $[4] = [7] = [16] = [-2]$ should all be in $\mathbb{Z}_3$. However, since they are all equivalent by Eqn. (2), actually **none** of them will be in $\mathbb{Z}_3$. We avoid this redundancy by only including classes where the representative is less than the value of n. For example, $\mathbb{Z}_3 = \{[0], [1], [2]\}$. Every integer (class) can then be mapped onto (to) a class within $\mathbb{Z}_3$.

Finally, addition and multiplication in $\mathbb{Z}_n$ are defined by $[a] \oplus [b] = [a+b]$ and $[a] \odot [b] = [ab]$, respectively. Tables 2 and 3 show the addition and multiplication Cayley tables for all elements of $\mathbb{Z}_3$. Note that, *e.g.*, in Table 2 $[2] \oplus [2] = [1]$, replacing $[4]$.

A further study can be done at this point and it is strongly encouraged that the reader explore congruence and modular arithmetic more deeply. However, for our purposes, this introduction suffices.

[6] When expressing congruence, the symbol '$\equiv$' must accompanied by '$(\bmod n)$' in order for it to actually *mean* anything. Without this, the same symbol of '$\equiv$' is used for definitions.

[7] We must emphasize what the congruence class is with respect to since, notice, $[5]$ by itself is insufficient information.
One may also arrive at Eqn. (2) by the definition of congruence in the previous paragraph. We know that $n \equiv a (\bmod c) \implies c|(n-a)$, which means $\exists k \in Z$ s.t. $n - a = kc$, hence $n = a + kc$.

[8] It is important to emphasize that this is a set of congruence classes, not integers. Each class itself is a set of integers, as seen in the previous paragraph.

$\oplus$	$[0]$	$[1]$	$[2]$
$[0]$	$[0]$	$[1]$	$[2]$
$[1]$	$[1]$	$[2]$	$[0]$
$[2]$	$[2]$	$[0]$	$[1]$

Table 2: Addition table for elements of $\mathbb{Z}_3$.

$\odot$	$[0]$	$[1]$	$[2]$
$[0]$	$[0]$	$[0]$	$[0]$
$[1]$	$[0]$	$[1]$	$[2]$
$[2]$	$[0]$	$[2]$	$[1]$

Table 3: Multiplication table for elements of $\mathbb{Z}_3$.

Question 7. *Make the Cayley tables for addition and multiplication in $\mathbb{Z}_4$. What do you notice about the column/row of $[2]$ under addition?*

Rings

Like groups, rings are best motivated by the generalization of properties of arithmetic in $\mathbb{Z}$ and $\mathbb{Z}_n$. Recall that in the Groups subsection we found that $(\mathbb{Z}, +)$ is an abelian group. Adding on to the 5 properties of abelian groups, we can list the following properties of the integers equipped with multiplication:

6. multiplication is closed, *i.e.*, the product of integers remains an integer;

7. multiplication is associative, *i.e.*, the *order in which we multiply* does not change the outcome;

8. the distributive properties hold;
9. multiplication is commutative, *i.e.*, the *order of the integers* being multiplied does not change the outcome;
10. a multiplicative identity exists, namely the integer 1, which can be multiplied to every integer without altering their value; and lastly
11. if the product of two integers is zero, then at least one of the original integers was itself zero.

Question 8. *Using Tables 2 and 3, verify that all of these properties hold for $\mathbb{Z}_3$.*

Question 9. *Do properties 6-11 hold for $\mathbb{Z}_n$, for any value of n? In particular, does property 11 hold for $\mathbb{Z}_4$? What about, $\mathbb{Z}_5$, $\mathbb{Z}_6$, and $\mathbb{Z}_7$? If you start to think the trend is parity-dependent, consider $\mathbb{Z}_2$. What must be true of n for properties 6-11 to hold for $\mathbb{Z}_n$?*

The last property in the list above is not something we often explore in elementary mathematics, maybe because it's trivial in the integers. This triviality will vanish as we move to more abstract settings. We begin the abstraction of these common properties with the following definition:

Because a ring is equipped with both addition and multiplication, we introduce the *distributive properties* to relate the two operations. These properties are axioms that must be satisfied in order for a set with two binary operations to be considered a ring.

Definition 10 (Ring)**.** A **ring** is an additive abelian group R equipped with multiplication, $(\cdot)$, s.t. $\forall\ a, b, c \in R$

6. $a \cdot b \in R$, (Closure under multiplication)
7. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$, (Associative multiplication)
8. $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$. (Distributive properties)

We use the symbol 0_R to denote the additive identity in a ring R. As of now, rings lack a multiplicative identity, denoted 1_R, and the existence of inverses for multiplication for them to form a group under multiplication as well. However, notice that rings introduce the distributive properties, which are not present in groups.

Rings which carry additional properties to the eight in Defn. 10 are given a different name. For example, a ring whose elements commute under multiplication is given a different name, though not as special as a commutative group:

Definition 11 (Commutative ring)**.** A ring R is a **commutative ring** if it satisfies the following axiom: $\forall\ a, b \in R$,

9. $ab = ba$. (Commutative multiplication)

If a ring forms both an additive abelian group and a multiplicative abelian group, we call it the following:

Definition 12 (Ring with identity)**.** A ring R which contains an element, say 1_R, satisfying the following axiom: $\forall\, a \in R$,

10. $a1_R = a = 1_R a$, (Multiplicative identity)

is a **ring with identity**.

Example 13. The set of integers $\mathbb{Z}$ equipped with the usual addition and multiplication operations, $(\mathbb{Z}, +, \cdot)$, is a commutative ring with identity. It satisfies all 10 properties listed in Defns. 10 through 12.

Example 14. Let E be the set of even integers. We claim that $(E, +, \cdot)$ is a commutative ring, but not a ring with identity. To see this, we verify that it satisfies properties 1-9 in Defns. 10 and 11.

1. If $a \in E$ and $b \in E$, then $a + b \in E$ since the sum of two even integers is even. That is, $a + b = 2m + 2n = 2(m + n)$ for some $m, n \in \mathbb{Z}$.
2. By the associativity of addition in $\mathbb{Z}$, $a + (b + c) = (a + b) + c$.
3. The additive identity in E is 0 since $\forall\, a \in E$, $a + 0 = a$. 0 is even since $0 = 2 \cdot 0$.
4. For any $a \in E$, the additive inverse is $-a$ since $a + (-a) = 0$. Note that $-a$ is even since $a = 2k$ for some $k \in \mathbb{Z} \implies -a = -2k = 2(-k)$.
5. By the commutativity of addition in $\mathbb{Z}$, $a + b = b + a$.
6. If $a \in E$ and $b \in E$, then $ab \in E$ since the product of two even integers is even. That is, $ab = (2m)(2n) = 2(2mn)$ for some $m, n \in \mathbb{Z}$.
7. By the associativity of multiplication in $\mathbb{Z}$, $a(bc) = (ab)c$.
8. By the distributive properties in $\mathbb{Z}$, $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.
9. By the commutativity of multiplication in $\mathbb{Z}$, $ab = ba$.

There is no multiplicative identity in E since $1 \notin E$. Thus, $(E, +, \cdot)$ is not a ring with identity.

Question 15. *Verify that the set of odd integers with the usual addition and multiplication is not a ring.*

Example 16. Consider the set of congruence class modulo n in $\mathbb{Z}$, $\mathbb{Z}_n$. We seek to show that $(\mathbb{Z}_n, \oplus, \odot)$ is a commutative ring with identity. To do so, we must verify that it satisfies all 10 properties listed in Defns. 10 through 12.

1. If $[a] \in \mathbb{Z}_n$ and $[b] \in \mathbb{Z}_n$, then $[a] \oplus [b] = [a+b] \in \mathbb{Z}_n$ by definition of addition in $\mathbb{Z}_n$.
2. By the definition of addition in $\mathbb{Z}_n$, $[a] \oplus ([b] \oplus [c]) = [a] \oplus [b+c] = [a+(b+c)] = [(a+b)+c] = [a+b] \oplus [c] = ([a] \oplus [b]) \oplus [c]$.
3. The additive identity in $\mathbb{Z}_n$ is $[0]$ since $\forall\ [a] \in \mathbb{Z}_n$, $[a] \oplus [0] = [a+0] = [a]$.
4. For any $[a] \in \mathbb{Z}_n$, the additive inverse is $[-a]$ since $[a] \oplus [-a] = [a+(-a)] = [a-a] = [0]$.
5. By definition of addition in $\mathbb{Z}_n$, $[a] \oplus [b] = [a+b] = [b+a] = [b] \oplus [a]$.
6. If $[a] \in \mathbb{Z}_n$ and $[b] \in \mathbb{Z}_n$, then $[a] \odot [b] = [ab] \in \mathbb{Z}_n$ by definition of multiplication in $\mathbb{Z}_n$.
7. By the definition of multiplication in $\mathbb{Z}_n$, $[a] \odot ([b] \odot [c]) = [a] \odot [bc] = [a(bc)] = [(ab)c] = [ab] \odot [c] = ([a] \odot [b]) \odot [c]$.
8. By the definition of addition and multiplication in $\mathbb{Z}_n$, $[a] \odot ([b] \oplus [c]) = [a] \odot [b+c] = [a(b+c)] = [ab+ac] = [ab] \oplus [ac] = ([a] \odot [b]) \oplus ([a] \odot [c])$. Similarly, $([a] \oplus [b]) \odot [c] = [ac] \oplus [bc]$.
9. By the definition of multiplication in $\mathbb{Z}_n$, $[a] \odot [b] = [ab] = [ba] = [b] \odot [a]$.
10. The multiplicative identity in $\mathbb{Z}_n$ is $[1]$ since $\forall\ [a] \in \mathbb{Z}_n$, $[a] \odot [1] = [a \cdot 1] = [a]$.

As an interesting note, suppose n is not prime. Then, there exists $a, b \in \mathbb{Z}$ s.t. $1 < a < n$, $1 < b < n$, and $ab = n$. Thus, in $\mathbb{Z}_n$, $[a] \odot [b] = [ab] = [n] = [0]$, even though neither $[a]$ nor $[b]$ is equal to $[0]$. Therefore, when n is not prime, $\mathbb{Z}_n$ contains what we call **zero divisors**, elements x and y such that $x \neq 0_R$, $y \neq 0_R$, but $xy = 0_R$. This property will be important later on when we study fields.

The study of rings often involves studying polynomials with coefficients in a given ring. For example, one may study polynomials with

integer coefficients. However, this is not something useful for our current development. As an alternative, we instead consider matrices.

Matrices

For the time being, we will not formally define matrices. Instead, we will provide an informal, very boring definition of what a matrix is, followed by some examples of matrix operations. A more formal definition will be provided later in the text when we study vector spaces.

Definition 17 (Matrix)**.** Let m and n be positive integers. An m-by-n **matrix A** is a rectagular array of elements arranged in m rows and n columns, denoted

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

The notation a_{ij} denotes the element in the i-th row and the j-th column of $\mathbf{A}$. In other words, the first index refers to the row and the second index refers to the column.

Notice that the definition says nothing about what the elements a_{ij} are. In theory, they can be elements from any set, it simply depends on what you want to achieve with the matrices. For our purposes, we will consider matrices with real number entries, denoted $\mathbf{M}(\mathbb{R})$.

Example 18. Let $\mathbf{M}(\mathbb{R})$ be the set of all 2×2 matrices with real number entries. That is, $\mathbf{M}(\mathbb{R})$ contains all arrays of the form

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where a, b, c, and d are real numbers. Two matrices are equal if and only if their corresponding entries are equal, *i.e.*,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \iff a = e,\ b = f,\ c = g,\ d = h.$$

For example,

$$\begin{pmatrix} 4 & 9 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1+3 & 4+5 \\ 7-7 & 8-6 \end{pmatrix}, \quad \text{but} \quad \begin{pmatrix} 4 & 9 \\ 0 & 2 \end{pmatrix} \neq \begin{pmatrix} 4 & 9 \\ 2 & 0 \end{pmatrix}.$$

Addition of matrices is done entry-wise:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1+5 & -2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 4 \\ 10 & 12 \end{pmatrix}.$$

Multiplication of matrices is a bit more involved. The product of two 2×2 matrices is given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 7 & 2 \\ 0 & 5 \end{pmatrix} \cdot \begin{pmatrix} 1 & 6 \\ 8 & 4 \end{pmatrix} = \begin{pmatrix} 7\cdot 1+2\cdot 8 & 7\cdot 6+2\cdot 4 \\ 0\cdot 1+5\cdot 8 & 0\cdot 6+5\cdot 4 \end{pmatrix} = \begin{pmatrix} 23 & 50 \\ 40 & 20 \end{pmatrix}.$$

Reversing the order of the factors in matrix multiplication *may* produce a different result, as is the case in our previous example:

$$\begin{pmatrix} 1 & 6 \\ 8 & 4 \end{pmatrix} \cdot \begin{pmatrix} 7 & 2 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 1\cdot 7+6\cdot 0 & 1\cdot 2+6\cdot 5 \\ 8\cdot 7+4\cdot 0 & 8\cdot 2+4\cdot 5 \end{pmatrix} = \begin{pmatrix} 7 & 32 \\ 56 & 48 \end{pmatrix}.$$

Multiplication is therefore not commutative in general for matrices. One can verify that $(\mathbf{M}(\mathbb{R}), +, \cdot)$ satisfies all the properties of a ring as given in Defn. 10 and is thus a ring. The zero element of said ring is the zero matrix

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

and the matrix $\mathbf{X} = \left(\begin{smallmatrix} -a & -b \\ -c & -d \end{smallmatrix}\right)$ is a solution of

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \mathbf{X} = \mathbf{0},$$

thus $\mathbf{X}$ is the additive inverse of $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The multiplicative identity element is the identity matrix

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

satisfying

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \mathbf{I} = \mathbf{I} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Interestingly, two nonzero matrices can multiply to give the zero matrix. For example,

$$\begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} -3 & -9 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} 4 \cdot (-3) + 6 \cdot 2 & 4 \cdot (-9) + 6 \cdot 6 \\ 2 \cdot (-3) + 3 \cdot 2 & 2 \cdot (-9) + 3 \cdot 6 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Using the same definitions of matrix addition and multiplication as above, we can construct other rings using matrices. Consider the following question:

Question 19. *Let* $\mathbf{K}(\mathbb{R})$ *be the set of all* 2×2 *matrices of the form*

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix},$$

where $a, b \in \mathbb{R}$. *Show that* $(\mathbf{K}(\mathbb{R}), +, \cdot)$ *is a commutative ring with identity.*

In the examples we've done so far, we've seen a couple of rings that have the property of two nonzero elements multiplying to give zero. In the following subsection, we explore rings that do not have this property. We also explore rings which have an additional, new property. Rings with these additional properties are given special names and are of incredible importance in the study of algebra.

Integral Domains and Fields

Definition 20 (Integral domain)**.** An **integral domain** is a commutative ring R with identity $1_R \neq 0_R$ that satisfies the following axiom: $\forall a, b \in R$,

11. if $ab = 0$, then $a = 0$ or $b = 0$.

The condition $1_R \neq 0_R$ is included here to avoid trivial cases where the ring contains only one element; namely, the ring $\{0_R\}$. Elements of a ring that are themselves nonzero but multiply to give zero are called **zero divisors**:

Definition 21 (Zero divisor)**.** An element a in a ring R is a **zero divisor** provided that $a \neq 0_R$ and $\exists\, b \in R$, with $b \neq 0_R$, s.t. $ab = 0_R$.

Looking back at Example 18, we see that $\mathbf{M}(\mathbb{R})$ is not an integral domain since it contains zero divisors; namely, the two nonzero matrices that we multiplied to get the zero matrix, $\left(\begin{smallmatrix} 4 & 6 \\ 2 & 3 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} -3 & -9 \\ 2 & 6 \end{smallmatrix}\right)$.

Recall our short dicussion in the section on groups, where we mentioned that $(\mathbb{Z}, \cdot)$ is not a group because not every integer has a multiplicative inverse. However, we pointed out that $(\mathbb{Q} \setminus \{0\}, \cdot)$ is a group because every nonzero rational number has a multiplicative inverse. Thus far in our development, we have not studied rings where every nonzero element has a multiplicative inverse. We do so now with the following definition:

Definition 22 (Field)**.** A **field** is a commutative ring R with identity $1_R \neq 0_R$ that satisfies the following axiom: $\forall\, a \in R$, with $a \neq 0_R$,

12. $\exists\, a^{-1} \in R$ s.t. $aa^{-1} = 1$.

Note axiom 11 is not mentioned explicitly in the definition of a field. However, it can be shown that every field is an integral domain.[9]

[9] The inverse of this is not true, however, as it is not generally true that an integral domain is a field. Take, for example, $\mathbb{Z}$. It is an integral domain but not a field. Interestingly, it is true that a *finite* integral domain is a field.

Example 23. The set of rational numbers $\mathbb{Q}$ equipped with the usual addition and multiplication operations, $(\mathbb{Q}, +, \cdot)$, is a field. It satisfies all 12 properties listed in Defns. 10 through 22. The set of real numbers $\mathbb{R}$ is also a field under the usual addition and multiplication.

Further Resources

1. *Abstract Algebra: An Introduction* (3rd ed.) by Thomas W. Hungerford.

The entirety of this `\section{}` is pulled from Hungerford's text. I strongly believe it is one of the best introductory texts on ring theory, and I highly recommend it to anyone looking to learn more about rings, integral domains, and fields.

Complex Numbers

Definitions and Algebraic Properties

Definition 24 (Complex numbers)**.** The set of **complex numbers**, denoted $\mathbb{C}$, is defined as pairs of real numbers,

$$\mathbb{C} \equiv \{(x, y) : x, y \in \mathbb{R}\},$$

equipped with addition

$$(x_1, y_1) + (x_2, y_2) \equiv (x_1 + x_2, y_1 + y_2),$$

and multiplication

$$(x_1, y_1) \cdot (x_2, y_2) \equiv (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2).$$

By the definition above complex numbers of the form $(x, 0)$ behave like real numbers under addition and multiplication:

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0), \quad \text{and} \quad (x_1, 0) \cdot (x_2, 0) = (x_1 x_2, 0).$$

Furthermore, the complex numbers $(0, 0)$ and $(1, 0)$ serve as the additive and multiplicative identities, respectively. With all of this, we now show that $(\mathbb{C}, +, \cdot)$ is a field:

Proposition 25. *The set of complex numbers $\mathbb{C}$ equipped with the addition and multiplication operations defined above, $(\mathbb{C}, +, \cdot)$, is a field.*

Question 26. *Prove the above proposition. (Answer provided below.)*

Proof. We must verify that $(\mathbb{C}, +, \cdot)$ satisfies all 11 properties listed in Defns. 10 through 22. $\forall$ (x_1, y_1), (x_2, y_2), $(x_3, y_3) \in \mathbb{C}$,

1. $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \in \mathbb{C}$ by closure of addition in $\mathbb{R}$,
2. $(x_1, y_1) + [(x_2, y_2) + (x_3, y_3)] = (x_1, y_1) + (x_2 + x_3, y_2 + y_3) = (x_1 + (x_2 + x_3), y_1 + (y_2 + y_3)) = ((x_1 + x_2) + x_3, (y_1 + y_2) + y_3) = (x_1 + x_2, y_1 + y_2) + (x_3, y_3) = [(x_1, y_1) + (x_2, y_2)] + (x_3, y_3)$ by associativity of addition in $\mathbb{R}$,
3. $\exists 0_F \in \mathbb{C}$, namely $(0, 0)$, s.t. $(x_1, y_1) + (0, 0) = (x_1 + 0, y_1 + 0) = (x_1, y_1) = (0 + x_1, 0 + y_1) = (0, 0) + (x_1, y_1)$ by the additive identity property in $\mathbb{R}$,
4. $\forall$ $(x_1, y_1) \in \mathbb{C}$, $\exists$ an additive inverse, namely $(-x_1, -y_1)$, s.t. $(x_1, y_1) + (-x_1, -y_1) = (x_1 - x_1, y_1 - y_1) = (0, 0) = (-x_1 + x_1, -y_1 + y_1) = (-x_1, -y_1) + (x_1, y_1)$ by the additive inverse property in $\mathbb{R}$,
5. $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) = (x_2 + x_1, y_2 + y_1) = (x_2, y_2) + (x_1, y_1)$ by commutativity of addition in $\mathbb{R}$,
6. $(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2) \in \mathbb{C}$ by closure of multiplication (and addition) in $\mathbb{R}$,

7. $(x_1, y_1) \cdot [(x_2, y_2) \cdot (x_3, y_3)] = (x_1, y_1) \cdot (x_2x_3 - y_2y_3, x_2y_3 + y_2x_3) = (x_1(x_2x_3 - y_2y_3) - y_1(x_2y_3 + y_2x_3), x_1(x_2y_3 + y_2x_3) + y_1(x_2x_3 - y_2y_3)) = (x_1x_2x_3 - x_1y_2y_3 - y_1x_2y_3 - y_1y_2x_3, x_1x_2y_3 + x_1y_2x_3 + y_1x_2x_3 - y_1y_2y_3) = (x_1x_2x_3 - y_1y_2x_3 - x_1y_2y_3 - y_1x_2y_3, x_1y_2x_3 + y_1x_2x_3 + x_1x_2y_3 - y_1y_2y_3) = ((x_1x_2 - y_1y_2)x_3 - (x_1y_2 + y_1x_2)y_3, (x_1y_2 + y_1x_2)x_3 + (x_1x_2 - y_1y_2)y_3) = (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2) \cdot (x_3, y_3) = [(x_1, y_1) \cdot (x_2, y_2)] \cdot (x_3, y_3)$ by associativity of multiplication and addition in $\mathbb{R}$,
8. $(x_1, y_1) \cdot [(x_2, y_2) + (x_3, y_3)] = (x_1, y_1) \cdot (x_2 + x_3, y_2 + y_3) = (x_1(x_2 + x_3) - y_1(y_2 + y_3), x_1(y_2 + y_3) + y_1(x_2 + x_3)) = (x_1x_2 + x_1x_3 - y_1y_2 - y_1y_3, x_1y_2 + x_1y_3 + y_1x_2 + y_1x_3) = ((x_1x_2 - y_1y_2) + (x_1x_3 - y_1y_3), (x_1y_2 + y_1x_2) + (x_1y_3 + y_1x_3)) = (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2) + (x_1x_3 - y_1y_3, x_1y_3 + y_1x_3) = (x_1, y_1) \cdot (x_2, y_2) + (x_1, y_1) \cdot (x_3, y_3)$. Similarly, $[(x_1, y_1) + (x_2, y_2)] \cdot (x_3, y_3) = (x_1, y_1) \cdot (x_3, y_3) + (x_2, y_2) \cdot (x_3, y_3)$ by associativity of multiplication and addition in $\mathbb{R}$,
9. $(x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2) = (x_2x_1 - y_2y_1, x_2y_1 + y_2x_1) = (x_2, y_2) \cdot (x_1, y_1)$ by commutativity of multiplication and addition in $\mathbb{R}$,
10. $\exists 1_F \in \mathbb{C}$, namely $(1, 0)$, s.t. $(x_1, y_1) \cdot (1, 0) = (x_1 \cdot 1 - y_1 \cdot 0, x_1 \cdot 0 + y_1 \cdot 1) = (x_1, y_1) = (1 \cdot x_1 - 0 \cdot y_1, 1 \cdot y_1 + 0 \cdot x_1) = (1, 0) \cdot (x_1, y_1)$ by the multiplicative identity property in $\mathbb{R}$,
11. $\forall\ (x_1, y_1) \in \mathbb{C}$, with $(x_1, y_1) \neq (0, 0)$, $\exists$ a multiplicative inverse, namely $\left(\frac{x_1}{x_1^2+y_1^2}, \frac{-y_1}{x_1^2+y_1^2}\right)$, s.t.

$$(x_1, y_1) \cdot \left(\frac{x_1}{x_1^2 + y_1^2}, \frac{-y_1}{x_1^2 + y_1^2}\right) = \left(\frac{x_1^2 + y_1^2}{x_1^2 + y_1^2}, \frac{0}{x_1^2 + y_1^2}\right) = (1, 0) = \left(\frac{x_1}{x_1^2 + y_1^2}, \frac{-y_1}{x_1^2 + y_1^2}\right) \cdot (x_1, y_1).$$

□

The multiplicative inverse used in the proof of property 11 above is derived from the fact that if we want

$$(x_1, y_1) \cdot (a, b) = (1, 0),$$

then we must have

$$(x_1a - y_1b, x_1b + y_1a) = (1, 0),$$

which gives us the system of equations

$$\begin{aligned} x_1a - y_1b &= 1, \\ x_1b + y_1a &= 0. \end{aligned}$$

Solving this system for a and b,

$$x_1b + y_1a = 0 \quad \Longrightarrow \quad b = \frac{-y_1a}{x_1} \quad \text{(assuming } x_1 \neq 0\text{)},$$

$$\therefore,\ x_1a - y_1\left(\frac{-y_1a}{x_1}\right) = 1 \quad \Longrightarrow \quad x_1a + \frac{y_1^2a}{x_1} = 1 \quad \Longrightarrow \quad \boxed{a = \frac{x_1}{x_1^2 + y_1^2}} \quad \text{and} \quad b = \frac{-y_1a}{x_1} = \boxed{\frac{-y_1}{x_1^2 + y_1^2}}.$$

The definition of multiplication in $\mathbb{C}$ may seem innocent at first glance, but it has deep implications. In particular, notice that

$$(0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0). \tag{3}$$

This identity, together with the fact that

$$(a, 0) \cdot (x, y) = (ax, ay),$$

allows for an alternative notation for complex numbers. Specifically, we can write

$$(x, y) = (x, 0) + (0, y) = (x, 0) \cdot (1, 0) + (y, 0) \cdot (0, 1),$$

where since $(x, 0)$ and $(y, 0)$ behave like real numbers, we can denote them simply as x and y, respectively. That is, we can write

$$(x, y) = x \cdot (1, 0) + y \cdot (0, 1).$$

This means that we can write any complex number (x, y) as a linear combination $(1, 0)$ and $(0, 1)$ with coefficients x and y. $(1, 0)$, in turn, behaves like the real number 1, while $(0, 1)$ behaves like a new number, call it i. Then the complex number that we used to call (x, y) can now be written as

$$(x, y) = x \cdot 1 + y \cdot i = x + yi.$$

Eqn. (3) then tells us that $i^2 = -1$.[10]

[10] $i^2 = -1$ reveals that i is the square root of -1. This not only says that an equation such as $x^2 + 1 = 0$ has a solution (or that the polynomial $x^2 + 1$ is *reducible*), but every nonconstant polynomial has roots in $\mathbb{C}$. This is the content of the Fundamental Theorem of Algebra, a deep and important result in mathematics that we will, unfortunately, ignore here.

Definition 27. The number x is called the **real part** and y the **imaginary part** of the complex number $x + yi$, often denoted $\mathrm{Re}\,(x + yi) = x$ and $\mathrm{Im}\,(x + yi) = y$, respectively.

Question 28. *Find the real and imaginary parts of $(x + iy)^4$, where, of course, $x, y \in \mathbb{R}$.* ***Hint:*** *Recall the binomial theorem, which states that*

$$(a + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k. \tag{4}$$

In particular, when $n = 4$, we have

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

Foreshadowing: some other useful expansions of Eqn. (4) are the $n = 2$ case:

$$(a + b)^2 = a^2 + 2ab + b^2,$$

and the $n = 3$ case:

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

Geometric Representation

Complex numbers are usually denoted by a single letter, such as z. Thus, we often write $z = x + yi$, where $x, y \in \mathbb{R}$. Because the product is commutative, there is no difference between writing $x + yi$ and $x + iy$. Visually, complex numbers can be represented as points in the Euclidean plane $\mathbb{R}^2$ having Cartesian coordinates (x, y). In this context we refer to $\mathbb{R}^2$ as the **complex plane** (alternatively, the

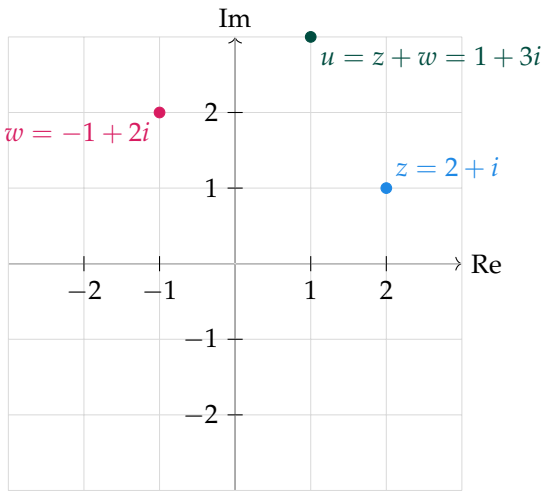


Figure 5: The complex plane. The horizontal axis represents the real part of a complex number, and the vertical axis represents the imaginary part. Here, we plot the complex numbers $z = 2 + i$, $w = -1 + 2i$, and their sum $u = z + w = 1 + 3i$. Notice that the sum of complex numbers is simply the sum of their corresponding coordinates in the plane.

Argand plane or the **Gaussian plane**).[11] We use the horizontal axis (the x-axis) to represent the real part of a complex number and the vertical axis (the y-axis) to represent the imaginary part.

In Fig. , we plot the complex numbers $z = 2 + i$ and $w = -1 + 2i$ in the complex plane. Notice that the sum of these two complex numbers, $u = z + w = 1 + 3i$, is represented by the point obtained by adding their corresponding coordinates in the plane. Going back to our original definition of complex numbers as pairs of real numbers, we see that this is precisely how addition is defined in $\mathbb{C}$:

$$(2,1) + (-1,2) = (2 + (-1), 1 + 2) = (1,3).$$

One may prefer our original notation of complex numbers as pairs of real numbers when performing addition. Doing so is particularly useful when considering complex numbers as straight arrows starting from the origin $(0,0)$ and ending at the point (x,y) in the complex plane. Fig. is the as in Fig. 6.

These arrows are more formally known as **vectors**. A vector in $\mathbb{R}^2$ is an ordered pair of real numbers that can be represented as an arrow in the plane. Vectors have both a magnitude (or length) and a direction. We define these notions for complex numbers as follows:

[11] Notice that the plane is defined as $\mathbb{R}^2$, not $\mathbb{C}^2$. The complex plane is a way of visualizing complex numbers, which are elements of $\mathbb{C}$, using points in $\mathbb{R}^2$.

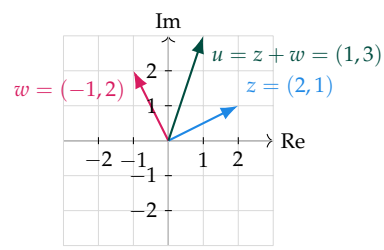


Figure 6: The complex plane with vectors representing complex numbers. Here, the complex numbers $z = (2,1)$, $w = (-1,2)$, and their sum $u = z + w = (1,3)$ are represented as arrows starting from the origin. The sum of complex numbers corresponds to the vector addition of their respective arrows.

Definition 29. The **modulus** (or **magnitude**) of a complex number $z = x + yi$ is

$$r = |z| \equiv \sqrt{x^2 + y^2},$$

and an **argument** (or **angle**) of z is any real number ϕ s.t.

$$x = r\cos(\phi) \quad \text{and} \quad y = r\sin(\phi).$$

By the above definition we can represent any complex number $z = x + yi$ in terms of its modulus r and an argument ϕ as

$$z = r\cos(\phi) + ir\sin(\phi).$$

The periodic nature of the trigonometric functions implies that if ϕ is an argument of z, then so is $\phi + 2\pi k$ for any integer k. Thus, every nonzero complex number has infinitely many arguments differing by integer multiples of 2π. The number $0 = 0 + 0i$ has modulus zero and every real number ϕ is an argument.

Example 30. The modulus of the complex number $z = 1 + i\sqrt{3}$ is given by

$$r = |z| = \sqrt{1^2 + \left(\sqrt{3}\right)^2} = \sqrt{4} = 2.$$

To find an argument ϕ of z, we solve the system of equations

$$\begin{aligned} 1 &= 2\cos(\phi), \\ \sqrt{3} &= 2\sin(\phi). \end{aligned}$$

Solving the first equation for $\cos(\phi)$ tells us that $\phi = \frac{\pi}{3}, \frac{5\pi}{3}$. Similarly, solving the second equation for $\sin(\phi)$ tells us that $\phi = \frac{\pi}{3}, \frac{2\pi}{3}$. The only value of ϕ that satisfies both equations is $\phi = \frac{\pi}{3}$. Thus, one argument of z is $\frac{\pi}{3}$, and all other arguments are given by

$$\phi = \frac{\pi}{3} + 2\pi k, \quad k \in \mathbb{Z}.$$

When the argument ϕ of a complex number z is restricted to the interval $(-\pi, \pi]$, it is called the **principal argument** of z and is denoted by $\arg(z)$.

Question 31. *Consider the complex number $z = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$. Find its modulus and principal argument. Then, compute the modulus and smallest nonnegative argument of the complex numbers z^2, z^3, and z^4. What do you observe?* ***Hint:*** *Refer to Eqn. (4) and the handy expansions!*

Example 32. Given the modulus $r = 3$ and an argument $\phi = \frac{\pi}{4}$, we can find the corresponding complex number z as follows:

$$z = r\cos(\phi) + ir\sin(\phi) = 3\cos\left(\frac{\pi}{4}\right) + 3i\sin\left(\frac{\pi}{4}\right) = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i.$$

Prior to proceeding, recall the following trigonometric identities:

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta), \tag{5}$$

$$\sin(\alpha + \beta) = \cos(\alpha)\sin(\beta) + \sin(\alpha)\cos(\beta). \tag{6}$$

Now, consider two complex numbers $z_1 = r_1\cos(\phi_1) + ir_1\sin(\phi_1)$ and $z_2 = r_2\cos(\phi_2) + ir_2\sin(\phi_2)$. Their product is given by

$$
\begin{aligned}
z_1 z_2 &= (r_1 \cos(\phi_1) + i r_1 \sin(\phi_1)) \cdot (r_2 \cos(\phi_2) + i r_2 \sin(\phi_2)) \\
&= r_1 r_2 \Big[\cos(\phi_1)\cos(\phi_2) + i\cos(\phi_1)\sin(\phi_2) + i\sin(\phi_1)\cos(\phi_2) + i^2 \sin(\phi_1)\sin(\phi_2)\Big] \\
&= r_1 r_2 [\cos(\phi_1)\cos(\phi_2) - \sin(\phi_1)\sin(\phi_2) + i\cos(\phi_1)\sin(\phi_2) + i\sin(\phi_1)\cos(\phi_2)] \\
&= r_1 r_2 [\cos(\phi_1)\cos(\phi_2) - \sin(\phi_1)\sin(\phi_2) + i(\cos(\phi_1)\sin(\phi_2) + \sin(\phi_1)\cos(\phi_2))] \\
&= r_1 r_2 [\cos(\phi_1 + \phi_2) + i\sin(\phi_1 + \phi_2)], \quad \text{by Eqns. (5), (6).}
\end{aligned}
\tag{7}
$$

Thus, the modulus of the product $z_1 z_2$ is the product of the moduli of z_1 and z_2, and an argument of $z_1 z_2$ is the sum of an argument of z_1 and an argument of z_2. In view of the above calculation, we will deal

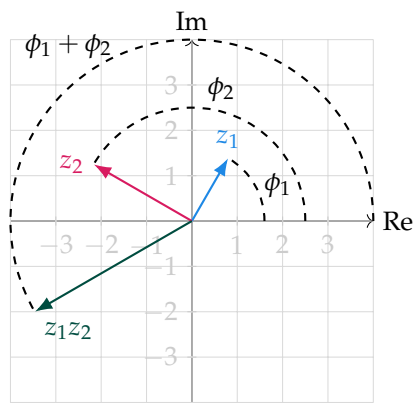


Figure 7: The complex plane with vectors representing complex numbers z_1, z_2, and their product $z_1 z_2$. The modulus of the product is the product of the moduli of z_1 and z_2, and an argument of the product is the sum of an argument of z_1 and an argument of z_2.

with complex numbers of the form $\cos(\phi) + i\sin(\phi)$, where ϕ is a real number. To simplify notation, we define

$$
e^{i\phi} \equiv \cos(\phi) + i\sin(\phi). \tag{8}
$$

At this point, this exponential notation is indeed purely a notation. However, later on we will see that this notation has deeper implications and connections to other areas of mathematics.

Fig. shows some examples of complex numbers expressed in this new notation. We demonstrate the usefulness of this notation with the following proposition.

Proposition 33. *For any ϕ, ϕ_1, $\phi_2 \in \mathbb{R}$,*

1. $e^{i\phi_1} e^{i\phi_2} = e^{i(\phi_1+\phi_2)}$,
2. $e^{i0} = 1$,
3. $\frac{1}{e^{i\phi}} = e^{-i\phi}$,
4. $e^{i(\phi+2\pi)} = e^{i\phi}$,
5. $\left|e^{i\phi}\right| = 1$,
6. $\frac{d}{d\phi} e^{i\phi} = i\, e^{i\phi}$.

Question 34. *Prove the above proposition. (Answer below.)*

Proof. Going one by one:

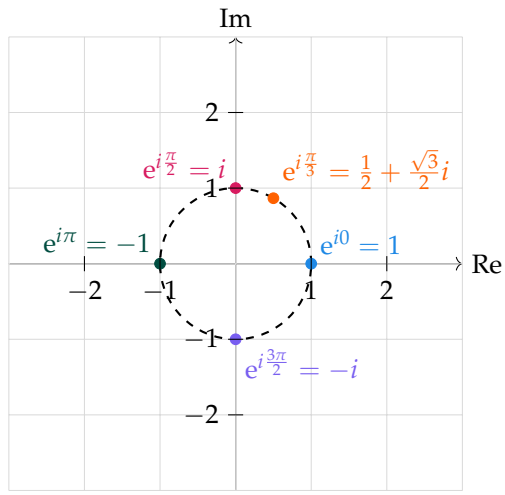


Figure 8: Examples of complex numbers in exponential notation on the complex plane. The resemblance to points on the unit circle is evident, as all these complex numbers have a modulus of 1.

1. Follows directly from Eqn. (7) with $r_1 = r_2 = 1$.
2. Follows directly from Eqn. (8) as
$$e^{i0} = \cos(0) + i\sin(0) = 1 + i \cdot 0 = 1.$$
3. To show that $\frac{1}{e^{i\phi}} = e^{-i\phi}$, we need to show that $e^{i\phi} \cdot e^{-i\phi} = 1$. This follows directly from property 1 and property 2 as
$$e^{i\phi} \cdot e^{-i\phi} = e^{i(\phi-\phi)} = e^{i0} = 1.$$
4. Follows directly from Eqn. (8) as
$$e^{i(\phi+2\pi)} = \cos(\phi + 2\pi) + i\sin(\phi + 2\pi) = \cos(\phi) + i\sin(\phi) = e^{i\phi},$$
by the periodicity of the trigonometric functions.
5. Follows directly from Eqn. (8) as
$$\left|e^{i\phi}\right| = \sqrt{\cos^2(\phi) + \sin^2(\phi)} = \sqrt{1} = 1,$$
by the definition of modulus in Defn. 29.
6. Follows directly from Eqn. (8) as
$$\frac{d}{d\phi} e^{i\phi} = \frac{d}{d\phi}[\cos(\phi) + i\sin(\phi)] = -\sin(\phi) + i\cos(\phi) = i[\cos(\phi) + i\sin(\phi)] = i\,e^{i\phi}.$$

□

With our new notation we can now express any complex number $z = x + yi$ in terms of its modulus r and an argument ϕ as
$$z = x + yi = r\,e^{i\phi}.$$

With this new representation we now have five different ways of representing a complex number z:

We conclude this `\section{}` with a proposition that follows the following definition:

Question 35. *Find the values of*
$$(1+2i)^3, \quad i^n, \quad (1+i)^n + (1-i)^n,$$
where $n \geqslant 0$.

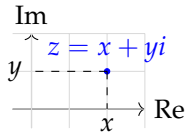
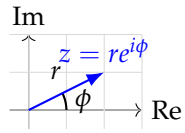
Formal (x, y)	Algebraic:	rectangular $x + iy$	exponential $re^{i\theta}$
	Geometric:	cartesian 	polar 

Table 4: Different representations of a complex number z.

Definition 36. The **(complex) conjugate** of a complex number $z = x + yi$ is the complex number $z^* = x - yi$. We denote the conjugate by

$$(x + yi)^* \equiv x - yi.$$

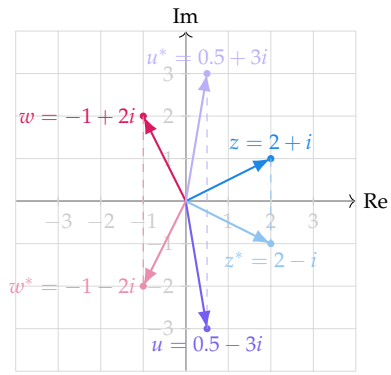


Figure 9: The complex plane with vectors representing complex numbers and their conjugates. Each conjugate is obtained by reflecting the original complex number w.r.t. the real axis.

Algebraically, all this is doing is changing the sign of the imaginary term. Geometrically, this is a reflection w.r.t. the real axis. This is shown in Fig. 9. We then see that the square of the modulus of a complex number $z = x + yi$ can be expressed as the product of z and its conjugate z^*:

$$|z|^2 = x^2 + y^2 = (x + yi)(x - yi) = zz^*. \tag{9}$$

The conjugate also enables us to consider division of complex numbers. Given two complex numbers $z_1 = x_1 + y_1 i$ and $z_2 = x_2 + y_2 i$ with $z_2 \neq 0$, their quotient is

$$\frac{z_1}{z_2} = \frac{x_1 + y_1 i}{x_2 + y_2 i}.$$

Similar to rationalizing the denominator when dealing with real numbers, we can multiply the numerator and denominator by the conjugate of the denominator to obtain

$$\frac{z_1}{z_2} = \frac{(x_1 + y_1 i)(x_2 - y_2 i)}{(x_2 + y_2 i)(x_2 - y_2 i)} = \frac{(x_1 x_2 + y_1 y_2) + i(y_1 x_2 - x_1 y_2)}{x_2^2 + y_2^2}.$$

This is now in the standard form $a + bi$ with

$$a = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2}, \quad b = \frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2}.$$

Equivalently, we can express the quotient in terms of moduli and arguments. If $z_1 = r_1 \, e^{i\phi_1}$ and $z_2 = r_2 \, e^{i\phi_2}$, then

$$\frac{z_1}{z_2} = \frac{r_1 \, e^{i\phi_1}}{r_2 \, e^{i\phi_2}} = \frac{r_1}{r_2} \, e^{i(\phi_1 - \phi_2)},$$

provided $r_2 \neq 0$. The following proposition collects additional useful properties of complex conjugation.

An alternative notation for the complex conjugate is $\bar{z}$. This notation is often used in various mathematical texts and contexts. Our use of the notation z^* is to hint towards its connection to the adjoint operation in linear algebra.

Proposition 37. *For any $z, z_1, z_2 \in \mathbb{C}$,*

1. $(z_1 \pm z_2)^* = {z_1}^* \pm {z_2}^*$,
2. $(z_1 \cdot z_2)^* = {z_1}^* \cdot {z_2}^*$,
3. $\left(\frac{z_1}{z_2}\right)^* = \frac{{z_1}^*}{{z_2}^*}$ *provided* $z_2 \neq 0$,
4. $(z^*)^* = z$,
5. $|z^*| = |z|$,
6. $|z|^2 = zz^*$,
7. $\mathrm{Re}\,(z) = \frac{1}{2}(z + z^*)$,
8. $\mathrm{Im}\,(z) = \frac{1}{2i}(z - z^*)$,
9. $\left(e^{i\phi}\right)^* = e^{-i\phi}$.

Question 38. *Prove the above proposition. (Answer below.)*

Proof. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Going one by one:

1. Follows directly from the definition of complex conjugation as $(z_1 \pm z_2)^* = ((x_1 + iy_1) \pm (x_2 + iy_2))^* = ((x_1 \pm x_2) + i(y_1 \pm y_2))^* = (x_1 \pm x_2) - i(y_1 \pm y_2) = (x_1 - iy_1) \pm (x_2 - iy_2) = {z_1}^* \pm {z_2}^*$.
2. Follows directly from the definition of complex conjugation as $(z_1 \cdot z_2)^* = ((x_1 + iy_1) \cdot (x_2 + iy_2))^* = ((x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2))^* = (x_1x_2 - y_1y_2) - i(x_1y_2 + y_1x_2) = (x_1 - iy_1) \cdot (x_2 - iy_2) = {z_1}^* \cdot {z_2}^*$.
3. Follows directly from the definition of complex conjugation as $\left(\frac{z_1}{z_2}\right)^* = \left(\frac{x_1+iy_1}{x_2+iy_2}\right)^* = \left(\frac{(x_1+iy_1)(x_2-iy_2)}{(x_2+iy_2)(x_2-iy_2)}\right)^* = \left(\frac{(x_1x_2+y_1y_2)+i(y_1x_2-x_1y_2)}{x_2^2+y_2^2}\right)^* = \frac{(x_1x_2+y_1y_2)-i(y_1x_2-x_1y_2)}{x_2^2+y_2^2} = \frac{(x_1x_2+y_1y_2)+i(x_1y_2-y_1x_2)}{x_2^2+y_2^2} = \frac{(x_1-iy_1)(x_2+iy_2)}{(x_2-iy_2)(x_2+iy_2)} = \frac{x_1-iy_1}{x_2-iy_2} = \frac{{z_1}^*}{{z_2}^*}$.
4. Follows directly from the definition of complex conjugation as $(z^*)^* = (x - iy)^* = x + iy = z$.
5. Follows directly from the definition of complex conjugation as $|z^*| = |x - iy| = \sqrt{x^2 + (-y)^2} = \sqrt{x^2 + y^2} = |z|$.
6. Shown in Eqn. (9).
7. Follows directly from the definition of complex conjugation as $\frac{1}{2}(z + z^*) = \frac{1}{2}((x + iy) + (x - iy)) = \frac{1}{2}(2x) = x = \mathrm{Re}\,(z)$.
8. Follows directly from the definition of complex conjugation as $\frac{1}{2i}(z - z^*) = \frac{1}{2i}((x + iy) - (x - iy)) = \frac{1}{2i}(2iy) = y = \mathrm{Im}\,(z)$.
9. Follows directly from Eqn. (8) as $\left(e^{i\phi}\right)^* = (\cos(\phi) + i\sin(\phi))^* = \cos(\phi) - i\sin(\phi) = \cos(-\phi) + i\sin(-\phi) = e^{-i\phi}$.

□

The advanced reader may note that properties 1, 2, and 3 imply that complex conjugation is a field automorphism of $\mathbb{C}$. That is, complex conjugation is a bijective map from $\mathbb{C}$ to itself that preserves both addition and multiplication.

Property 6. implicitly states the inverse of a nonzero complex number z as

$$z^{-1} = \frac{1}{z} = \frac{z^*}{|z|^2}.$$

Further Resources

1. *A First Course in Complex Analysis* by Beck, M., et al. Version 1.54. https://matthbeck.github.io/complex.html

This entire `\section{}` comes from Beck's brilliantly intuitive text. It is highly recommended for anyone looking to deepen their understanding of complex analysis. One considering further study in complex analysis should definitely consider this text. For a more traditional approach, consider the following standard undergraduate texts:

2. *Complex Analysis* by Ahlfors, L. V. McGraw-Hill, 1979.
3. *Complex Variables and Applications* by Brown, J. W., and Churchill, R. V. McGraw-Hill, 2009.

If not directly taken from Ahlfors, the questions in this `\section{}` are inspired by exercises from these texts.

Linear Algebra

Vectors: Algebraic and Geometric Interpretations

Definition 39. A **list of length *n*** (or an ***n*-tuple**), where $n \in \mathbb{Z}_{\geqslant 0}$, is an ordered collection of n elements separated by commas and surrounded by parentheses:

$$(x_1, x_2, \ldots, x_n).$$

Two **lists are equal** iff they have the same length and the same elements are in the same order.

In general, the elements of a list can be numbers, other lists, or even more abstract objects. A list of length 0 is called the **empty list** and is denoted by $()$. Additionally, by definition, a list cannot have infinite length.

Definition 40. Let F be a field. $\boldsymbol{F^n}$ is the set of all lists of length n whose elements are in F:

$$F^n \equiv \{(x_1, x_2, \ldots, x_n) : x_i \in F \text{ for } i = 1, 2, \ldots, n\}.$$

For $(x_1, x_2, \ldots, x_n) \in F^n$ and $j \in \{1, 2, \ldots, n\}$, we say that x_j is the ***j*-th coordinate** of the list.

In particular, $\mathbb{R}^n$ is the set of all lists of length n whose elements are real numbers, and $\mathbb{C}^n$ is the set of all lists of length n whose

elements are complex numbers. We saw in the previous section that we initially treated the elements of $\mathbb{C}$ as ordered pairs of real numbers. Addition of lists in F^n is defined exactly as addition of ordered pairs in $\mathbb{C}$: consider $(x_1, x_2, \ldots, x_n)$ and $(y_1, y_2, \ldots, y_n)$, with $x_i, y_i \in F$ for all $i = 1, 2, \ldots, n$, their sum is

$$(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) \equiv (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n). \quad (10)$$

We denote the 0 list in F^n by

$$0_{F^n} \equiv (0_F, 0_F, \ldots, 0_F),$$

where 0_F is the additive identity in F. Under this addition operation, we propose the following:

Proposition 41. *Let F be a field. The set F^n, equipped with addition as defined in Eqn. (10), is an additive abelian group.*

Question 42. *Prove the above proposition. (Answer below.)*

Proof. We are tasked with showing that F^n with addition satisfies the group axioms (Defn. 3) and is abelian (Defn. 5). Going one by one, for all $(x_1, x_2, \ldots, x_n)$, $(y_1, y_2, \ldots, y_n)$, and $(z_1, z_2, \ldots, z_n)$ in F^n:

1. The sum $(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$ is also in F^n since $x_i + y_i \in F$ for all $i = 1, 2, \ldots, n$ (by the closure property of fields).
2. We have

$$\begin{aligned}
(x_1, x_2, \ldots, x_n) + ((y_1, y_2, \ldots, y_n) + (z_1, z_2, \ldots, z_n)) &= (x_1, x_2, \ldots, x_n) + (y_1 + z_1, y_2 + z_2, \ldots, y_n + z_n) \\
&= (x_1 + (y_1 + z_1), x_2 + (y_2 + z_2), \ldots, x_n + (y_n + z_n)) \\
&= ((x_1 + y_1) + z_1, (x_2 + y_2) + z_2, \ldots, (x_n + y_n) + z_n) \\
&= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) + (z_1, z_2, \ldots, z_n) \\
&= ((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) + (z_1, z_2, \ldots, z_n)
\end{aligned}$$

   where the third equality follows from the associativity property of fields.
3. The 0 list $0_{F^n} = (0_F, 0_F, \ldots, 0_F)$ satisfies

$$(x_1, x_2, \ldots, x_n) + 0_{F^n} = (x_1 + 0_F, x_2 + 0_F, \ldots, x_n + 0_F) = (x_1, x_2, \ldots, x_n)$$

   where the second equality follows from the identity property of fields. Similarly, $0_{F^n} + (x_1, x_2, \ldots, x_n) = (x_1, x_2, \ldots, x_n)$.
4. The additive inverse of $(x_1, x_2, \ldots, x_n)$ is $(-x_1, -x_2, \ldots, -x_n)$ since

$$(x_1, x_2, \ldots, x_n) + (-x_1, -x_2, \ldots, -x_n) = (x_1 + (-x_1), x_2 + (-x_2), \ldots, x_n + (-x_n)) = (0_F, 0_F, \ldots, 0_F) = 0_{F^n},$$

   where the second equality follows from the inverse property of fields. Similarly, $(-x_1, -x_2, \ldots, -x_n) + (x_1, x_2, \ldots, x_n) = 0_{F^n}$.

5. We have

$$\begin{aligned}(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) &= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) \\ &= (y_1 + x_1, y_2 + x_2, \ldots, y_n + x_n) \\ &= (y_1, y_2, \ldots, y_n) + (x_1, x_2, \ldots, x_n),\end{aligned}$$

where the second equality follows from the commutativity property of fields.

Therefore, F^n with addition is an additive abelian group.

□

In proving property 4 in the above proposition, we see that the additive inverse of a list $(x_1, x_2, \ldots, x_n)$ is obtained by negating each of its coordinates, giving $(-x_1, -x_2, \ldots, -x_n)$. Thus, there is a *global* negative sign that can be "factored" out:

$$-(x_1, x_2, \ldots, x_n) \equiv (-x_1, -x_2, \ldots, -x_n).$$

However, there is nothing special about a negative one that makes it different from any other value. We can easily define multiplication of a list by *any* scalar[12] $a \in F$:

$$a(x_1, x_2, \ldots, x_n) \equiv (ax_1, ax_2, \ldots, ax_n).$$

[12] A **scalar** is simply an element of the field F.

More formally, we have the following definition:

Definition 43. The **product** of a value $a \in F$ and a list $(x_1, x_2, \ldots, x_n) \in F^n$ is defined by multiplying each coordinate of the list by a:

$$a(x_1, x_2, \ldots, x_n) \equiv (ax_1, ax_2, \ldots, ax_n).$$

This is what we call **scalar multiplication**.

Loosely speaking, lists that satisfy the above addition and scalar multiplication operations are called (row) **vectors**.[13] We are ommitting a formal definition of vectors here, as we will soon introduce the more general concept of vector spaces. For the time being, we seek to get a better understanding of the geometry of lists in F^n under addition and scalar multiplication.

[13] Vectors are written in **boldface** with an arrow over them, $\vec{\mathbf{v}}$. The term **vector** has multiple meanings in mathematics. In this context, we are referring to lists that satisfy specific algebraic properties under addition and scalar multiplication.

We've seen before that the complex plane uses $\mathbb{R}^2$ to represent complex numbers. More generally, we can use F^n to represent points (or vectors) in an n-dimensional space over a field F. For example, $\mathbb{R}^3$ is used to represent points (or vectors) in three-dimensional Euclidean space. As before, when we think of a list as an arrow, we refer to it as a vector. For instance, the vector $(2, 3)$ in $\mathbb{R}^2$ can be

represented as an arrow from the origin $(0,0)$ to the point $(2,3)$ in the Cartesian plane. Similarly, the vector $(1,4,2)$ in $\mathbb{R}^3$ can be represented as an arrow from the origin $(0,0,0)$ to the point $(1,4,2)$ in three-dimensional space. In the following figure we show examples of vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ and illustrate how their components lie on the axes.

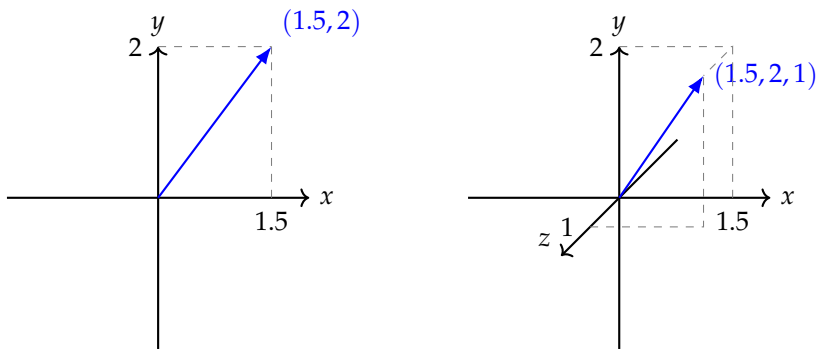


Figure 10: Showing vectors in $\mathbb{R}^2$ (left) and $\mathbb{R}^3$ (right) along with their components on each axis. There is an inherent order in the structure of lists, which is reflected in the labeling of the axes as x, y, and z. The first component of each vector corresponds to the x-axis, the second to the y-axis, and the third (if applicable) to the z-axis.

Visually, the addition of vectors corresponds to the *parallelogram law*. The parallelogram law for vector addition states that the sym of two vectors $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ that act at the same point P is the vector beginning at P that is represented by the diagonal of the parallelogram having $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ as adjacent sides. The point P is often taken to be the origin. This is illustrated in the following figure:

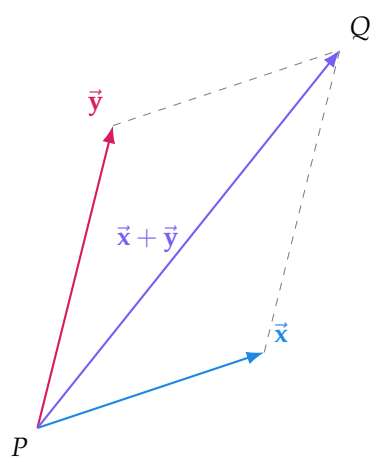


Figure 11: Since opposite sides of a parallelogram are parallel and of equal length, the endpoint Q of the arrow representing $\vec{\mathbf{x}} + \vec{\mathbf{y}}$ can be found by either allowing $\vec{\mathbf{x}}$ to act at P and then allowing $\vec{\mathbf{y}}$ to act at the endpoint of $\vec{\mathbf{x}}$, or by allowing $\vec{\mathbf{y}}$ to act at P and then allowing $\vec{\mathbf{x}}$ to act at the endpoint of $\vec{\mathbf{y}}$. Thus two vectors that both act at the point P may be added "tail-to-head," or "tip-to-tail."

Finally, scalar multiplication of a vector $\vec{\mathbf{v}}$ by a scalar a can be interpreted geometrically as stretching or compressing the vector $\vec{\mathbf{v}}$ by a factor of $|a|$ if $a > 0$, or reflecting the vector $\vec{\mathbf{v}}$ across the origin and then stretching or compressing it by a factor of $|a|$ if $a < 0$. If $a = 0$, the resulting vector is the zero vector, **not** the value 0. This is

in accordance with Defn. 43. Each of these geometric interpretations can be visualized in the following figure:

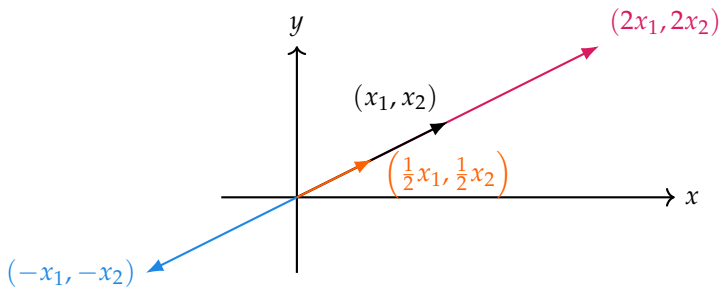


Figure 12: Scalar multiplication on a vector may affect its length or direction. The original vector is shown here in black. Multiplying by a scalar greater than 1 (red) stretches the vector, multiplying by a scalar between 0 and 1 (orange) compresses the vector, and multiplying by a negative scalar (blue) reflects the vector across the origin. Multiplying the vector by 0 would result in the zero vector at the origin (not shown).

Two nonzero vectors $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ are called **parallel** if $\vec{\mathbf{y}} = a\vec{\mathbf{x}}$ for some nonzero $a \in \mathbb{R}$. In particular, if $a < 0$, then $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ point in opposite directions and are then called **antiparallel**.

Question 44. *This question consists of four parts.*

1. *Find $x \in \mathbb{R}^4$ s.t.*

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

2. *Find the midpoint of the line segment joining the points (a, b) and (c, d) in $\mathbb{R}^2$.*

Let $\vec{\mathbf{v}}_1 = (a, 0)$ and $\vec{\mathbf{v}}_2 = (b, c)$ in $\mathbb{R}^2$. See Fig. 13 for the following questions.

3. *Find the area of the parallelogram formed by the vectors $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$. You may assume, without loss of generality, that a, b, and c are all positive.*

4. *We see in Fig. 13 that the vector $\vec{\mathbf{v}}_1 + \vec{\mathbf{v}}_2$ corresponds to one of the diagonals of the parallelogram formed by $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$. Find an expression, in terms of $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$, for the vector corresponding to the other diagonal of the parallelogram.*

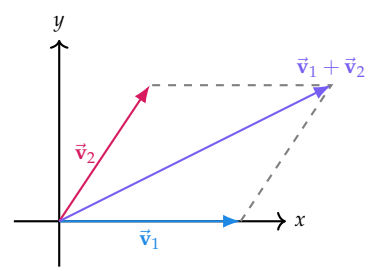


Figure 13: Reference figure for parts 3–4 of Quest. 44. The parallelogram is formed by the vectors $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$ in $\mathbb{R}^2$.

(Finite Dimensional) Vector Spaces

We've now arrived at the thing we've spent so long building up to:

Definition 45 (Vector space)**.** Let F be a field. A **vector space over F** is an additive abelian group (*i.e.*, an abelian group equipped with addition) V equipped with scalar multiplication s.t., $\forall\ a, a_1, a_2 \in F$ and $v, v_1, v_2 \in V$,

1. $a(v_1 + v_2) = av_1 + av_2$,

2. $(a_1 + a_2)v = a_1 v + a_2 v$,

3. $a_1(a_2 v) = (a_1 a_2)v$,

4. $1_F v = v$,

where 1_F is the multiplicative identity in F.

With this definition, we define our new use of the word **vector**:

Definition 46 (Vector)**.** Elements of a vector space are called **vectors** (or **points**).

The reader should not confuse this use of the word "vector" with the physical entity we discussed at the end of the previous `\subsection{}`.[14]

A vector space over $\mathbb{R}$ is called a **real vector space**, while a vector space over $\mathbb{C}$ is called a **complex vector space**. The set F^n, equipped with the addition and scalar multiplication operations defined in the previous `\subsection{}`, is a vector space over F. Though we were quite informal in our treatment of lists in the previous `\subsection{}`, vectors in F^n may be written as **column vectors**:

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

rather than as **row vectors**:

$$\begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}.$$

Suppose V is a vector space over a field F and that w and v_1, v_2, …, v_n are elements of V. We say that w is a **linear combination** of v_1, v_2, …, v_n if w can be written in the form

$$w = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

for $a_i \in F$. If $a_1 = a_2 = \cdots = a_n = 0_F$, we have the **trivial representation** of the zero vector, $w = 0_V$, as a linear combination of v_1, v_2, …, v_n. In general, all possible linear combinations of a set of vectors form the following definition:

[14] One should have also noticed by now that the notation for vectors in this section is different from that used in the previous subsection. We no longer use boldface with an arrow over it to denote vectors, as we are now working in a more abstract setting. In fact, vectors in a vector space need not even have a geometric interpretation at all!

Example 47. In $\mathbb{R}^3$, $(17, -4, 2)$ is a linear combination of $(2, 1, -3)$ and $(1, -2, 4)$ because

$$\begin{pmatrix} 17 \\ -4 \\ 2 \end{pmatrix} = 6\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + 5\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}.$$

Definition 48 (Span)**.** The set of all linear combinations of a set of vectors $v_1, v_2, \ldots, v_n$ in a vector space V over a field F is called the **span** of $v_1, v_2, \ldots, v_n$, denoted by $\operatorname{span}(v_1, v_2, \ldots, v_n)$. In symbols,

$$\operatorname{span}(v_1, v_2, \ldots, v_n) \equiv \{a_1 v_1 + a_2 v_2 + \cdots + a_n v_n : a_i \in F\}.$$

If every element of said vector space V is a linear combination of $v_1, v_2, \ldots, v_n$, we say that the set $\{v_1, v_2, \ldots, v_n\}$ **spans V** over F.

Example 49. In $\mathbb{R}^3$, the set

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

spans $\mathbb{R}^3$ because any vector

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$$

can be written as a linear combination of these three vectors:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

These three vectors are called the standard basis vectors of $\mathbb{R}^3$ and are often represented as simply $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ respectively, implying that each vector corresponds to an axis in $\mathbb{R}^3$.

Example 50. In $\mathbb{R}^2$, the set of a single vector

$$\left\{ \begin{pmatrix} a \\ b \end{pmatrix} \right\}$$

does not span $\mathbb{R}^2$ since, *e.g.*, the vector

$$\begin{pmatrix} a+2 \\ b+1 \end{pmatrix}$$

cannot **generally** be written as a linear combination of $\begin{pmatrix} a \\ b \end{pmatrix}$. Linear combinations of $\begin{pmatrix} a \\ b \end{pmatrix}$ are of the form

$$c \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ca \\ cb \end{pmatrix},$$

for some $c \in \mathbb{R}$. That is, the set $\left\{ \left(\begin{smallmatrix} a \\ b \end{smallmatrix} \right) \right\}$ only spans the line through the origin and the point (a, b) in $\mathbb{R}^2$.

Question 51. *In Exmp. 50 we said that the vector* $\left(\begin{smallmatrix} a+2 \\ b+1 \end{smallmatrix} \right)$ *cannot* ***generally*** *be written as a linear combination of* $\left(\begin{smallmatrix} a \\ b \end{smallmatrix} \right)$. *Provide specific values of* a *and* b *s.t. this is this case, i.e.,*

$$\begin{pmatrix} a+2 \\ b+1 \end{pmatrix} \neq c \begin{pmatrix} a \\ b \end{pmatrix},$$

for any $c \in \mathbb{R}$. *On the other hand, provide expressions for* a *and* b *in terms of* c *s.t. the following holds:*

$$\begin{pmatrix} a+2 \\ b+1 \end{pmatrix} = c \begin{pmatrix} a \\ b \end{pmatrix}.$$

Definition 52 (Linear dependence)**.** A subset S of a vector space V over a field F is said to be **linearly dependent** over F if there exist a finite number of distinct vectors $v_1, v_2, \ldots, v_n$ in S and scalars $f_1, f_2, \ldots, f_n$ in F, not all zero, s.t.

$$f_1 v_1 + f_2 v_2 + \cdots + f_n v_n = 0_V,$$

with each $f_i \in F$, then, $\forall i$, $f_i = 0_F$. A set that is not linearly dependent is said to be **linearly independent**.

The following facts about linearly indpendent sets are true in any vector space:

1. The empty set is linearly independent, for linearly dependent sets must be nonempty.
2. A set containing of a single nonzero vector is linearly independent. If $\{v\}$ is linearly dependent, then there exists $f \in F$, $f \neq 0_F$, s.t. $fv = 0_V$. Multiplying both sides by f^{-1} gives $v = 0_V$. Thus, $\{v\}$ is linearly dependent only if $v = 0_V$.
3. A set is linearly independent iff the only representations of the zero vector as a linear combination of its elements is the trivial representation.

The third fact provides a useful method for determining whether a (finite) set is linearly independent.[15]

A linearly independent spanning set for a vector space possesses an incredibly useful property: every vector in said vector space can be uniquely expressed as a linear combination of the vectors in the set.[16] It is this property that makes linearly independent spanning sets the building blocks of vector spaces.

Definition 53 (Basis)**.** A subset $\{v_1, v_2, \ldots, v_n\}$ of a vector space V over a field F is said to be a **basis** of V if it spans V and is linearly independent over F.

Example 54. In F^n, let $e_1 = (1, 0, \ldots, 0)$, $e_2 = (0, 1, \ldots, 0)$, $\ldots$, $e_n = (0, 0, \ldots, 1)$. Then $\{e_1, e_2, \ldots, e_n\}$ is a linearly independent spanning set for F^n, and thus forms a basis for F^n. As we saw in Example 49, these vectors are the standard basis vectors for F^n.

[15] This is done by setting up a system of linear equations and checking if the only solution is the trivial one. Traditionally, this involves writing the vectors as columns of a matrix and performing row reduction to echelon form. If this means nothing to you, don't worry; it **won't** be explained in more detail later because we won't need it for our purposes.

[16] The uniqueness of this representation is a direct consequence of linear independence and can be shown by assuming two different representations and subtracting them to obtain a nontrivial linear combination of the basis vectors that equals zero, which contradicts linear independence.

Definition 55 (Dimensionality)**.** If a vector space V over a field F has a finite basis, then V is said to be **finite dimensional** over F. The **dimension of V over F** is the number of elements in *any* basis of V. If V does not have a finite basis, then V is said to be **infinite dimensional** over F.

Lemma 56. *Let V be a vector space over a field F. The subset $\{v_1, v_2, \ldots, v_n\}$ of V is linearly dependent over F iff some v_k is a linear combination of v_1, v_2, …, v_{k-1}.*

Proof. If some v_k is a linear combination of other elements in V, then the set is linearly dependent by Defn. **??**. Conversely, suppose $\{v_1, v_2, \ldots, v_n\}$ is linearly dependent. Then $\exists\, f_1, \ldots, f_n \in F$, not all zero, s.t. $f_1v_1 + f_2v_2 + \cdots + f_nv_n = 0_V$. Let k be the largest index s.t. f_k is nonzero. Then $f_i = 0_F$ for $i > k$ and

$$\begin{aligned} f_1v_1 + f_2v_2 + \cdots + f_kv_k &= 0_V \\ f_kv_k &= -f_1v_1 - f_2v_2 - \cdots - f_{k-1}v_{k-1}. \end{aligned}$$

Since F is a field and $f_k \neq 0_F$, f_k^{-1} exists. Multiplying the preceding equation by f_k^{-1}, we have

$$v_k = -f_k^{-1}f_1v_1 - f_k^{-1}f_2v_2 - \cdots - f_k^{-1}f_{k-1}v_{k-1},$$

showing that v_k is a linear combination of the preceding v's. □

Lemma 57. *Let V be a vector space over a field F that is spanned by the set $\{v_1, v_2, \ldots, v_n\}$. If $\{u_1, u_2, \ldots, u_m\}$ is any linearly independent subset of V, then $m \leqslant n$.*

Resources

1. *Abstract Algebra: An Introduction* (3rd ed.) by Thomas W. Hungerford.

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